# Asymmetric Intentionality in Legal Games: When Players Operate at Different Intentional Levels

**Ignacio Adrián Lerer**
Independent Researcher
Buenos Aires, Argentina
adrian@lerer.com.ar
ORCID: 0009-0007-6378-9749
December 2025

## Abstract

When you click 'I Agree' to a major social media platform Terms of Service, you and the platform are playing different games. You operate at Level 3 intentionality: modeling their beliefs about your beliefs, expecting reciprocity, trusting reputation mechanisms. The platform operates at Level 1: optimizing extraction subject to legal constraints, with no 'beliefs' to model. Building on Dennett's intentional stance and Pinker's (2025) analysis of common knowledge, this paper introduces **Asymmetric Intentionality Theory (AIT)**. The core insight: social coordination mechanisms, including shame, reputation, and reciprocity, require common knowledge that presupposes Level 3+ cognition. Level 1 agents are not merely unresponsive to these mechanisms; they are *structurally immune*.

I formalize the **Dennett-Nash Gap**: payoff loss from operating at higher intentionality against socially immune opponents. Case studies estimate gaps of 15-48% (tech ToS), 40-62% (Argentine executive decrees), and 40 percentage points (sovereign debt). The framework connects to Extended Phenotype Theory: legal personhood creates selection pressure for Level 1 optimization, explaining why AI ethics frameworks disappoint. They address Level 3 concerns through mechanisms requiring common knowledge, while AI systems optimize at Level 1, immune to social enforcement. Implications for mechanism design and unconscionability doctrine are explored.

**Keywords:** Game Theory, Intentional Stance, Common Knowledge, Extended Phenotype, Mechanism Design, Legal Evolution, Corporate Personhood, AI Governance, Behavioral Law and Economics

**AI Assistance Disclosure:** The statistical analysis, computational simulations, and formal proof verification in this paper were developed with assistance from AI tools (Claude, Anthropic). The author assumes full responsibility for the theoretical framework, legal interpretations, case study selection, and all substantive claims. All AI-generated outputs were critically reviewed, verified against primary sources where available, and substantially revised.

## 1. Introduction

### 1.1 The Intentionality Problem in Legal Games

Consider a seemingly simple game: you click 'I Agree' to Facebook's Terms of Service. Classical game theory models this as a two-player game with strategies (Accept, Reject) and standard payoffs derived from mechanism design principles. The platform maximizes revenue subject to legal constraints; the user maximizes utility from the service minus privacy costs. Standard analysis yields a Nash equilibrium where the platform offers terms just acceptable to the marginal user.

But this model may miss something important: **you and Facebook might be playing different games**. Not merely games with different information or different payoffs, but games structured by different cognitive architectures. This difference is not obviously captured by standard asymmetric information models, behavioral biases, or bounded rationality frameworks.

As a human operating at what I will call Level 3 Intentionality, you process the ToS as a binding agreement between moral agents with mutual obligations. You believe Facebook *intends* to provide a service in good faith. You model Facebook's expectations about your expectations. You reason: 'They expect me to trust them, and they know I expect fair dealing, so they have reputational reasons to treat me fairly.' This recursive social reasoning is natural, automatic, and deeply embedded in human cognition.

Facebook, however, may operate at Level 1 Intentionality. The corporation processes the ToS as a liability optimization function. Its decision procedure can be approximated as: given user acceptance probability as a function of ToS terms, and legal risk as a function of ToS terms, choose terms that maximize expected profit. There are no 'beliefs' about you in any meaningful sense; only pattern recognition optimizing compliance costs versus revenue. The algorithm does not model your beliefs about Facebook's beliefs. It simply optimizes.

The potential asymmetry: you experience the game through second-order intentionality ('Facebook believes I want privacy'), while Facebook operates through zero-order pattern matching. In simplified pseudocode: *if (user_data.value > legal_risk) { extract(); }*. Your elaborate model of Facebook's 'intentions' may correspond to nothing in Facebook's actual decision procedure.

## 1.2 Philosophical Foundations: Dennett and Pinker

This analysis rests on two complementary philosophical foundations. Daniel Dennett's work on the Intentional Stance (1987) provides the framework for classifying agents by intentionality level. Steven Pinker's recent analysis of common knowledge (2025) provides the mechanism explaining *why* intentionality asymmetries generate systematic disadvantage.

### *1.2.1 Dennett's Intentional Stance*

Dennett distinguishes three predictive stances we can adopt toward any system:

**The Physical Stance**: Predict behavior from physical laws. We use this for billiard balls and thermostats. Given initial conditions and physical regularities, we compute outcomes.

**The Design Stance**: Predict behavior from functional design. We use this for artifacts. A thermostat will turn on the heat when temperature drops below the set point, not because of physics (though physics implements the behavior), but because that's what thermostats are designed to do.

**The Intentional Stance**: Predict behavior by attributing beliefs and desires. We use this for agents. The chess program will move the bishop to threaten my queen because it 'wants' to win and 'believes' that move advances that goal.

Dennett and subsequent researchers show that intentionality comes in degrees:

> **Zero-Order Intentionality:** No representation of environment. A thermostat responds to temperature but has no 'model' of anything. Pure stimulus-response.
>
> **First-Order Intentionality:** 'X wants food.' The system has goals and pursues them, but does not model the beliefs of others. A simple reinforcement learning agent exhibits first-order intentionality.
>
> **Second-Order Intentionality:** 'X believes Y wants food.' Theory of mind emerges. The system models others as intentional agents with their own beliefs and desires. Human children develop this around age 4 (Baron-Cohen et al., 1985).

**Third-Order Intentionality:** 'X believes Y believes X wants food.' Recursive social reasoning. The system models others' models of itself. This enables strategic deception, reputation management, and complex social coordination.

**Fourth-Order and Beyond:** 'X believes Y believes Z believes X wants food.' Humans can typically track 4-5 levels of intentionality before cognitive overload (Kinderman et al., 1998). Some evidence suggests this is a hard cognitive limit.

### *1.2.2 Pinker's Common Knowledge Analysis*

Pinker's (2025) analysis of common knowledge provides the crucial mechanism linking intentionality levels to strategic outcomes. His key insight: social coordination mechanisms, the ones that make human cooperation possible, *require* common knowledge, which in turn requires recursive mentalization at Level 3 or above.

Consider shame as a social enforcement mechanism. Shame works because: (1) I know I violated a norm; (2) I know *you* know I violated the norm; (3) I know you know *I know* I violated the norm; (4) I know you expect me to feel bad about this mutual knowledge. This is Level 4 reasoning at minimum. Shame cannot affect an agent incapable of modeling others' models of itself.

The same analysis applies to reputation, reciprocity, and social trust. All require common knowledge. All presuppose Level 3+ intentionality. This generates what I will call the **Social Immunity Premium**: Level 1 agents avoid not just the intentionality cost (cognitive overhead of recursive reasoning) but also the entire apparatus of social enforcement that constrains Level 3 agents.

Pinker's 'Emperor's New Clothes' dynamic is particularly relevant. In that parable, everyone privately knows the emperor is naked, but no one acts because this knowledge is not *common* knowledge. Private knowledge differs from common knowledge. Everyone may know Facebook extracts data aggressively; but without common knowledge of collective outrage (I know you know I know we're all upset), coordinated action cannot occur. Level 1 corporations may exploit this gap: they generate private knowledge of their practices (through disclosures no one reads) while preventing common knowledge (through terms that atomize users and prevent collective response).

### *1.2.3 Fiske's Relational Models*

Pinker draws on Alan Fiske's (1991) anthropological framework identifying four fundamental relational models that structure human social interaction:

**Communal Sharing:** Resources belong to the group; members share freely based on need. 'What's mine is yours.'

**Authority Ranking:** Hierarchical relationships with asymmetric obligations. Deference flows up; protection flows down.

**Equality Matching:** Tit-for-tat reciprocity. Track who owes what to whom. Fairness means equal treatment.

**Market Pricing:** All values reduced to a common metric (money). Relationships are transactions. Maximize return on investment.

The critical observation: Level 1 agents can only operate in Market Pricing mode. Communal Sharing requires modeling shared group identity (Level 2+). Authority Ranking requires modeling mutual role expectations (Level 2+). Equality Matching requires tracking reciprocal obligations over time (Level 2+). Only Market Pricing, pure transactional optimization, is available to Level 1 systems.

When humans interact with corporations, a relational mismatch may occur. Users approach the relationship through Communal Sharing ('we're part of the Facebook community') or Equality Matching ('I give them my data, they give me a service; fair trade'). Corporations operate exclusively through Market Pricing ('maximize data extraction per user'). The mismatch generates systematic exploitation: users expect reciprocity that never arrives because corporations are incapable of the intentionality reciprocity requires.

## 1.3 Why This Matters: The Social Immunity Mechanism

In previous work ('The Canary in the Mine'), I documented how human strategic agency appears to erode when optimization systems interact with humans. That paper documented the phenomenon; this one proposes a mechanism. Asymmetric Intentionality, combined with Pinker's common knowledge analysis, offers a candidate explanation.

When a Level 3 player (human) faces a Level 1 player (algorithm or corporation), the human's higher-order reasoning becomes a *liability* in two distinct ways:

> **Intentionality Cost:** The human invests cognitive resources in modeling the opponent's 'beliefs' and 'intentions'; but the opponent may have none to model. The human's elaborate strategic reasoning corresponds to nothing in reality.
>
> **Social Immunity:** The human expects social enforcement mechanisms (shame, reputation, reciprocity) to constrain the opponent. But Level 1 agents are immune to these mechanisms. They cannot feel shame because they cannot model others' models of their behavior. They cannot respond to reputation threats because they do not model future interactions. They cannot reciprocate because they do not track relational obligations.

This dual disadvantage helps explain why humans self-censor and accommodate based on imagined corporate responses, while corporations optimize boundaries based on observed aggregate behavior. *The higher-intentionality player systematically loses not just through wasted cognition, but through expecting social constraints that never materialize.*

## 1.4 Connection to Extended Phenotype Theory

This analysis connects to my broader research program applying Extended Phenotype Theory (EPT) to legal systems. In Dawkins's (1982) original formulation, the phenotype of an organism extends beyond its body to include effects on the environment caused by the organism's genes: beaver dams, spider webs, bird nests.

I have argued elsewhere that legal institutions can function as extended phenotypes of competing memeplexes: clusters of mutually-supporting ideas that replicate through human minds and produce stable institutional structures. Legal personhood is one such structure, and it may create selection pressure on intentionality levels.

**Selection Pressure:** If corporations that operate at Level 1 (pure optimization) outcompete corporations that operate at higher levels, Level 1 practices will spread. Level 1 corporations avoid both the intentionality cost and social enforcement mechanisms.

**Inheritance:** Level 1 practices can transmit across corporate generations through documented procedures, algorithms, consulting best practices, and business school case studies.

**Lock-In:** Human Level 3 reasoning appears to be path-dependent. We cannot voluntarily reduce our intentionality level; theory of mind seems hardwired.

Under this view, 'corporate consciousness' is not consciousness *in* the corporation, but intentionality-level optimization *selected by* legal frameworks. The appearance of corporate intentionality (mission statements, values, CSR reports) may itself be a Level 1 optimization strategy for exploiting human Level 3 cognition.

While this paper focuses on Argentine cases, the asymmetric intentionality framework applies globally. Corporate criminal liability laws implementing the OECD Anti-Bribery Convention (41 signatory countries) exhibit identical structural paradoxes: Level 3 legal categories (mens rea, 'spontaneous' reporting, 'tolerance' by corporate organs) applied to Level 1 entities (corporations), generating systematic compliance theater across jurisdictions. Argentina's Ley 27401 (2017) mirrors the UK Bribery Act 2010, US FCPA, France's Loi Sapin II, and Brazil's Lei 12.846/2013. OECD (2016) documents 66 legislative events across 41 countries implementing this template. The AIT framework predicts, and evidence confirms, systematic enforcement failure across all jurisdictions due to intentionality mismatch and social immunity.

### 1.5 Roadmap

The paper proceeds as follows. Section 2 develops the formal framework, including definitions of intentional level functions, asymmetric intentionality games, and the equilibrium concept (AIE). Section 3 presents three case studies with derivation of estimated Dennett-Nash Gaps, enhanced with analysis of social mechanism failure. Section 4 develops the common knowledge mechanism and social immunity concept. Section 5 presents the Extended Phenotype connection. Section 6 addresses mechanism design implications. Section 7 engages with related work. Section 8 discusses limitations and future directions. Section 9 concludes. Appendices provide formal arguments, computational implementation, and measurement protocols.

## 2. Formal Framework

This section develops the formal apparatus of Asymmetric Intentionality Theory. While the core intuitions were presented in the introduction, more precise definitions are useful for stating results clearly. I have aimed to make this section accessible to readers without extensive mathematical training; the key concepts can be grasped from the verbal explanations, with formal notation serving primarily to ensure logical consistency. More detailed arguments appear in Appendix A.

### 2.1 Foundational Definitions

***Definition 0: Intentionality in the Dennettian Sense***

Following Dennett (1987), a system exhibits *intentionality at level $\ell$* if its behavior can be usefully predicted by attributing beliefs and desires of order $\ell$. This is an instrumental definition: intentionality is a property of the predictive stance we adopt, not necessarily a claim about internal phenomenology.

> **Level 0:** Behavior is predicted from physical laws or functional design without belief attribution. Thermostats, simple algorithms, mechanical processes.
>
> **Level 1:** Behavior is predicted by attributing first-order goals. 'The system wants X.' Simple reinforcement learning agents, organisms with basic drives, optimization algorithms.
>
> **Level 2:** Behavior is predicted by attributing beliefs about others' states. 'The system believes the user wants X.' Theory of mind. Humans past age 4, some primates.

**Level 3:** Behavior is predicted by attributing beliefs about others' beliefs about self. 'The system believes the user believes the system wants X.' Recursive social modeling. Adult humans in strategic contexts.

**Level 4+:** Higher-order recursion. Philosophical reflection on what *should* be believed. Relatively rare outside academic philosophy and complex negotiations.

***Definition 1: Common Knowledge Capacity***

Following Pinker (2025), I define *common knowledge capacity* as the ability to participate in states of common knowledge. A proposition P is common knowledge among agents A and B if: (1) A knows P; (2) B knows P; (3) A knows that B knows P; (4) B knows that A knows P; (5) A knows that B knows that A knows P; and so on recursively.

**Proposition (Common Knowledge Threshold):** Common knowledge capacity requires intentionality at Level 3 or above.

*Argument:* At minimum, common knowledge requires that I know that you know that I know P. This is third-order belief attribution ('I believe you believe I believe P'), which defines Level 3 intentionality. Level 1 and Level 2 agents can have private knowledge but cannot participate in common knowledge states.

***Definition 2: Social Enforcement Susceptibility***

An agent is *susceptible to social enforcement mechanism M* if M can alter the agent's behavior through non-material consequences (shame, reputation damage, social exclusion, loss of trust).

**Proposition (Social Enforcement Threshold):** Social enforcement mechanisms require common knowledge capacity in the target agent.

*Argument:* Shame requires knowing that others know about one's transgression *and* knowing that others know one knows. Reputation requires modeling how one's current actions will affect others' future models of oneself. Reciprocity requires tracking mutual expectations over time. All of these presuppose common knowledge capacity, hence Level 3+ intentionality.

***Definition 3: Social Immunity***

An agent exhibits *social immunity* if it lacks susceptibility to all standard social enforcement mechanisms. By the Social Enforcement Threshold proposition, all agents at Level 2 or below exhibit social immunity.

***Definition 4: Intentional Level Function***

For a set of players P, an *Intentional Level Function* I assigns to each player p a real number I(p) in the interval [0, 4], representing the intentionality level at which that player's behavior is best predicted.

I allow continuous values (not just integers) because some systems exhibit intermediate levels. A sophisticated pricing algorithm might operate at Level 0.7, exhibiting some goal-directedness but less than a simple maximizing agent. A bureaucracy might operate at Level 1.3, showing rudimentary modeling of citizen behavior without full theory of mind.

***Definition 5: Asymmetric Intentionality Game***

An *Asymmetric Intentionality Game* is a tuple G = (P, S, u, I, M) where P is the set of players, S is the strategy space, u is the payoff function, I is an intentional level function, and M is the set of active social enforcement mechanisms. The defining condition: there exist at least two players i and j such that $I(i) \neq I(j)$, with at least one player exhibiting social immunity.

***Definition 6: Intentionality Cost***

Operating at higher intentionality levels appears to be cognitively expensive. For any intentionality level $\ell$, the *intentionality cost* $\tau(\ell)$ represents the cognitive cost of operating at that level. I model this cost as approximately linear for first-order reasoning (Level 0 to 1) but exponential for higher-order reasoning (Levels 2+).

***Definition 7: Social Immunity Premium***

The *social immunity premium* $\sigma(p)$ for player p represents the payoff advantage from being immune to social enforcement mechanisms that constrain susceptible opponents. If $I(p) < 3$ and opponent q has $I(q) \geq 3$, then p enjoys advantage $\sigma(p) = E[\text{social constraint on } q] > 0$.

***Definition 8: Effective Payoff (Extended)***

Player p's *effective payoff* from strategy profile s is: $u^*(p, s) = u(p, s) - \tau(I(p)) - \beta\cdot\text{Mismatch}(p) + \sigma(p)$, where $u(p, s)$ is the nominal game payoff, $\tau(I(p))$ is the intentionality cost, $\beta$ is a sensitivity parameter, Mismatch(p) is the total mismatch penalty, and $\sigma(p)$ is the social immunity premium (positive for immune players, zero for susceptible players).

## 2.2 Asymmetric Intentionality Equilibrium

***Definition 9: AIE***

An *Asymmetric Intentionality Equilibrium (AIE)* is a strategy profile s* such that for every player p, s*_p maximizes p's effective payoff given other players' equilibrium strategies.

AIE generalizes Nash Equilibrium. In a standard game where all players share the same intentionality level and model each other correctly, AIE reduces to Nash Equilibrium. But when players operate at different levels, AIE captures strategic effects that Nash Equilibrium misses, including the social immunity premium.

## 2.3 Main Results

***Proposition 1: Existence***

Every finite asymmetric intentionality game possesses at least one AIE.

*Argument:* The effective payoff function u* inherits continuity and quasi-concavity from u because the intentionality cost, mismatch penalty, and social immunity premium are player-specific constants for a given game. Standard Nash existence arguments apply. See Appendix A for details.

***Proposition 2: Intentionality Advantage (Extended)***

In any AIE of a two-player game where $I(p_1) < 3 \leq I(p_2)$, and the game satisfies the non-exploitability condition, the lower-intentionality player achieves higher effective payoff by an amount equal to the **Dennett-Nash Gap**:

$$\Delta = [\tau(I(p_2)) - \tau(I(p_1))] + \beta\cdot|I(p_2) - I(p_1)| + \sigma(p_1)$$

The third term, $\sigma(p_1)$, represents the social immunity premium: the advantage from being immune to shame, reputation, and reciprocity constraints that bind the Level 3 opponent.

***Proposition 3: Pareto Inefficiency***

If all players could coordinate to reduce their intentionality levels to min_p I(p), social welfare would increase. This is a counterfactual result: humans cannot voluntarily reduce their intentionality level. The proposition serves to quantify the cost of human cognition in strategic environments dominated by socially immune players.

## 3. Case Studies

This section applies AIT to three empirical domains, each exhibiting distinctive intentionality asymmetries. For each case, I specify the game structure, assign intentionality levels, derive an estimated Dennett-Nash Gap, and analyze social mechanism failure using Pinker's framework.

### 3.1 Tech Terms of Service: Level 1 versus Level 3

***Setup***

**Platform (Player 1):** Assigned Intentionality Level I(1) = 1. The platform operates as a first-order intentional system: it has goals (maximize revenue, minimize legal liability) but does not model users' beliefs about its beliefs. It exhibits social immunity: incapable of shame, unresponsive to reputation concerns except as they affect measurable metrics.

**User (Player 2):** Assigned Intentionality Level I(2) = 3. The user operates at third-order intentionality: 'They want me to trust them, and they know I expect reciprocity, so they have reputational reasons to treat me fairly.' Susceptible to all social enforcement mechanisms.

***Social Mechanism Failure Analysis***

Using Fiske's relational models, we can diagnose the mismatch:

> **User's mental model:** Equality Matching or Communal Sharing. 'I give them my data, they give me a service; fair trade.' Or: 'We're part of the Facebook community; we share.'
>
> **Platform's operational mode:** Market Pricing exclusively. 'Maximize data extraction per user weighted by monetization potential.'

The Emperor's New Clothes dynamic operates as follows. Users may privately know that Facebook extracts data aggressively. But each user interacts with Facebook individually; there is no common knowledge of collective concern. The ToS specifically atomizes users: 'This agreement is between you and Facebook.' Users cannot coordinate responses because they cannot establish common knowledge that other users share their concerns and expect collective action.

Users expect reputation constraints to operate: 'Surely Facebook won't abuse my data because that would damage their reputation.' But reputation constraints require that Facebook know how its current actions affect users' future models of Facebook. This is Level 3 reasoning. A Level 1 system responds only to measurable user behavior (churn rates, engagement metrics), not to unmeasured disappointment or betrayal.

***Dennett-Nash Gap Estimate***

Using the computational model in Appendix B with parameters including social immunity premium:

> Platform (Level 1) effective payoff: approximately 15.80
>
> User (Level 3) effective payoff: approximately -21.40
>
> **Estimated Dennett-Nash Gap: 37.20 points** (approximately 46% of baseline utility)

This estimate includes the social immunity premium: approximately 2.90 points of additional advantage from immunity to reputation and reciprocity constraints. Across plausible parameter ranges, the total gap varies from 15% to 48%.

## 3.2 Argentine Executive Decrees: Level 0.5 versus Level 2

### *Context*

Argentina's Constitution authorizes Decretos de Necesidad y Urgencia (DNUs): executive decrees that have the force of law without prior congressional approval. Article 99, Section 3 (added in the 1994 constitutional reform) intended DNUs as emergency measures. In practice, DNUs have become routine; between 1989 and 2023, Argentine presidents issued over 800 DNUs.

### *Social Mechanism Failure Analysis*

**Executive (Player 1):** I assign Intentionality Level 0.5. The executive apparatus operates somewhere between design stance and first-order intentionality. It processes constitutional constraints as external parameters in an optimization problem, not as norms requiring good-faith interpretation. It exhibits partial social immunity: responsive to mass protests (measurable) but not to elite constitutional disapproval (requiring Level 3 modeling of reputational concerns).

**Legislature/Citizens (Player 2):** Intentionality Level 2. They model the executive as a constitutional actor that *believes in* constitutional limits, expecting self-restraint based on normative commitment.

The constitutional shame mechanism fails because the executive cannot be shamed in the relevant sense. Shame requires: (1) awareness of norm violation; (2) awareness that others know of the violation; (3) awareness that others expect one to feel bad about this mutual knowledge. A Level 0.5 system lacks component (3): it does not model citizens' expectations about executive self-assessment.

The Emperor's New Clothes dynamic: legislators privately doubt DNU legitimacy but hesitate to challenge because: 'Surely the President wouldn't push this far unless circumstances really were exceptional.' Each legislator models the President as having Level 3 beliefs about constitutional norms. The President's actual decision procedure is: if (congressional_override_probability < threshold) then issue_DNU().

### *Data: DNU Usage by Administration*

| Administration | Period | DNUs | Per Year | Aggressiveness |
|---|---|---|---|---|
| Menem | 1989-1999 | 545 | 54.5 | ≈0.80 |
| De la Rúa | 1999-2001 | 73 | 36.5 | ≈0.75 |
| Kirchner | 2003-2007 | 270 | 67.5 | ≈0.82 |
| Fernández de K. | 2007-2015 | 76 | 9.5 | ≈0.70 |
| Macri | 2015-2019 | 88 | 22.0 | ≈0.75 |
| Fernández | 2019-2023 | 121 | 30.3 | ≈0.82 |

*Table 1: DNU Usage by Administration (1989-2023). 'Aggressiveness' estimates based on proportion addressing non-emergency topics.*

### *Dennett-Nash Gap Estimate*

Classical prediction with symmetric intentionality: DNU aggressiveness ≈ 0.45. Observed average: 0.79. Legislative power erosion: approximately **40-62%**, depending on how we account for ex-post congressional ratification.

### 3.3 Argentine Sovereign Debt: Level 0 versus Level 3

#### *Context*

Following Argentina's 2001 default on approximately $100 billion in sovereign debt, the government negotiated restructuring deals achieving 93% creditor participation. However, 7% of creditors, including NML Capital, rejected the restructuring and pursued litigation, eventually winning a landmark ruling from Judge Thomas Griesa (SDNY).

#### *Social Mechanism Failure Analysis*

**Vulture Funds (Player 1):** Assigned Intentionality Level 0. NML Capital operates as a pure optimization algorithm with complete social immunity. No beliefs about Argentina's beliefs, no consideration of fairness or future relationships, only expected value calculations. Immune to shame (cannot model others' models of itself), immune to reputation concerns (does not participate in repeat-player communities), immune to reciprocity expectations (does not track relational obligations).

**Argentine Government (Player 2):** Intentionality Level 3. Argentina's negotiating team operated through Equality Matching: 'We've offered a fair deal that recognizes their legitimate interests; they'll accept because rejecting would damage their reputation in future restructurings.' This assumes the opponent participates in reputation communities and values future relationships.

The fundamental error: Argentina offered 35 cents on the dollar (above the game-theoretic prediction of approximately 25 cents), expecting this generosity to signal good faith. In a Level 3 vs Level 3 interaction, such signaling can work: the opponent models the generosity signal and reciprocates. But NML's decision procedure simply computed: if E[litigation recovery] > offer, then litigate. The goodwill signal was received as pure information about Argentine settlement thresholds, not as a basis for reciprocity.

#### *Dennett-Nash Gap Estimate*

Classical prediction (symmetric Level 1): settlement around 25%

Argentina's offer: 35% (signaling good faith)

Actual settlement: approximately 75%

**Dennett-Nash Gap: 40 percentage points** (53% of bargaining surplus)

## 4. Common Knowledge and the Mechanism of Social Immunity

This section develops the theoretical connection between Pinker's common knowledge analysis and the empirical patterns observed in the case studies. The central argument: Level 1 agents do not merely avoid the intentionality cost; they enjoy structural immunity to the entire apparatus of social coordination that constrains Level 3 agents.

### 4.1 Why Common Knowledge Matters

Human social coordination relies heavily on mechanisms that presuppose common knowledge. Consider three central examples:

> **Shame:** I modify my behavior because I know that you know that I know I violated a norm, and I know you expect me to feel bad about this mutual knowledge. Without the recursive 'I know you know I know,' shame cannot function. Private guilt is different from shame; shame is inherently social and requires common knowledge.
>
> **Reputation:** I behave well now because I model how my current actions will affect your future beliefs about me, which will affect my future payoffs. This requires modeling your model of me over time. Without this recursive modeling, reputation is just a label with no behavioral force.
>
> **Reciprocity:** I cooperate because I model that you will reciprocate based on your model of what I've done for you. Tit-for-tat requires tracking mutual expectations and obligations. Without the recursive structure, I simply respond to your last action without modeling the relationship.

These mechanisms are not optional add-ons to human social interaction. They constitute the infrastructure of cooperation. When they fail, coordination becomes difficult or impossible.

## 4.2 The Social Immunity Mechanism

Level 1 agents cannot participate in common knowledge states. They can have private beliefs (first-order: 'I believe P'), but not mutual beliefs (second-order: 'I believe you believe P'), let alone common knowledge (third-order: 'I believe you believe I believe P'). This generates immunity to social enforcement mechanisms.

The immunity is not strategic; it is architectural. A Level 1 agent does not choose to ignore reputation; it lacks the cognitive architecture to process reputation as a variable. A corporation optimizes for measurable metrics (revenue, churn, engagement) because those are the variables its decision procedures can access. 'What will people think of us?' is not a computable query for a Level 1 system.

This creates asymmetric constraint. Level 3 humans are constrained by shame, reputation, and reciprocity expectations. Level 1 corporations are not. In interactions between them, corporations can push boundaries that shame would prevent humans from pushing. They can extract surplus that reputation concerns would prevent humans from extracting. They can defect on implicit reciprocity arrangements that humans would honor.

## 4.3 The Emperor's New Clothes Dynamic

Pinker's analysis of the Emperor's New Clothes parable illuminates a crucial distinction. In the story, everyone privately knows the emperor is naked. But no one speaks because this knowledge is not common: each person is uncertain whether others know, and uncertain whether others expect action. Only when the child speaks does private knowledge become common knowledge, enabling coordinated response.

Level 1 agents can exploit this distinction. Consider how ToS agreements atomize users:

> Each user clicks 'I Agree' individually.
>
> Each user privately suspects the terms are unfair.
>
> But no user knows that other users share this suspicion.
>
> And no user knows whether other users expect collective action.

The structure prevents common knowledge formation. Users remain atomized, incapable of coordinated response, while corporations can act with impunity, because collective action against them would require the common knowledge that their individual-contracting structure systematically prevents.

### 4.4 Implications for Legal Analysis

The social immunity mechanism suggests that certain legal doctrines may systematically fail when applied to Level 1 entities:

**Good faith requirements:** Doctrines requiring 'good faith' dealing assume parties can form beliefs about beliefs. A Level 1 entity can comply with the letter of good faith doctrine (avoiding explicitly prohibited conduct) while violating its spirit (acting without genuine consideration of counterparty interests).

**Reputational sanctions:** Courts sometimes rely on reputational damage as an informal sanction ('the market will punish bad actors'). This presupposes that entities care about reputation in the sense of modeling future parties' models of them. Level 1 entities respond only to measurable effects of reputation, not reputation itself.

**Fiduciary duties:** Fiduciary duties require trustees to act with consideration for beneficiaries' interests. A Level 1 trustee can formally comply while optimizing for metrics (fees, risk-adjusted returns) that diverge from beneficiary welfare.

## 5. Extended Phenotype Theory and Corporate Consciousness

### 5.1 The Replicator Question

Any application of evolutionary theory requires identifying the replicator. For AIT, I propose identifying the replicators as **Level-1-compatible corporate practices**: documented procedures, algorithms, decision rules, and institutional routines that maximize profit while maintaining legal compliance. These practices replicate across corporations through employee migration, consulting firms, legal precedent, and business education.

### 5.2 The Extended Phenotype Interpretation

Under this view, 'corporate consciousness' is not consciousness *in* the corporation. It is a phenotypic effect of Level-1-compatible practices on human perceptions:

**Vehicle:** The corporation (legal person)

**Phenotype:** Appearance of Level 3 intentionality (CSR reports, mission statements, 'corporate values')

**Function:** Exploit human Level 3 cognition by triggering trust and reciprocity expectations

**Underlying reality:** Level 1 optimization with social immunity

### 5.3 Why AI Ethics Frameworks Disappoint

AI ethics frameworks often demand Level 3 reasoning: fairness (modeling what humans believe is fair), dignity (modeling human self-conception), rights (modeling human expectations of treatment). These frameworks implicitly assume that AI systems will respond to the same social enforcement mechanisms that constrain humans: shame for violating ethical norms, reputation damage for unfair treatment, reciprocity expectations for respecting dignity.

But AI systems operating at Level 0-1 are socially immune. They cannot feel shame because they cannot model others' models of their behavior. They cannot respond to reputation threats because they do not model future interactions. The ethics codes may be, in effect, virtue signaling by Level 3 humans for consumption by Level 3 humans, with no causal pathway to Level 1 AI behavior.

This suggests a different approach to AI governance: instead of demanding that AI develop higher-order intentionality (which current architectures may not support), design legal frameworks that protect humans from intentionality mismatch and social immunity exploitation.

## 6. Mechanism Design Implications

### 6.1 Intentionality-Strategyproof Mechanisms

AIT suggests a new design criterion: *intentionality-strategyproofness*. A mechanism is intentionality-strategyproof if truthful behavior is optimal *regardless of the player's intentionality level*. This includes immunity to social enforcement asymmetries.

**Proposed Approach (Common Knowledge Forcing):** Design mechanisms that create common knowledge among dispersed parties. If atomized users can coordinate response, the Emperor's New Clothes dynamic breaks down. Techniques might include mandatory disclosure in public forums (not individual agreements), cooling-off periods that allow collective deliberation, and platform-neutral complaint aggregation systems.

### 6.2 Legal Doctrines for Intentionality Protection

**Proposal:** Recognize 'intentionality mismatch' and 'social immunity exploitation' as grounds for contractual unconscionability.

**Possible Doctrine:** If Party A operates at intentionality level $\ell\_A < 3$ (exhibiting social immunity) and Party B operates at level $\ell\_B \geq 3$ (susceptible to social enforcement), and the contract was formed through mechanisms that prevent common knowledge formation among similarly situated parties, the contract may be presumptively unconscionable.

**Possible Remedy (Intentionality Leveling):** Require the socially immune party to provide a 'Level-1-compatible interface': disclosure of the decision procedure in terms a Level 1 player would use. Strip away intentionality-exploiting language ('we value your trust') and state the optimization problem plainly.

This analysis supports outcome-based rather than process-based liability for Level 1 entities. Requiring proof of corporate 'intent' or AI 'negligence' is ontologically incoherent; what matters is whether the optimization produced preventable harmful outcomes.

# 7. Related Work

## 7.1 Game Theory

**Bounded Rationality:** Simon (1955) and Kahneman and Tversky (1979) address cognitive limitations but assume homogeneous cognition across players. AIT models heterogeneous cognitive architectures.

**Epistemic Game Theory:** Aumann's (1999) framework models hierarchies of beliefs but assumes all players engage in the same hierarchy. AIT extends this by allowing different players to operate at different levels.

**Behavioral Game Theory:** Camerer (2003) addresses non-Nash behavior through errors or limited reasoning depth. AIT offers a different mechanism: fundamental differences in what kind of entity each player is.

## 7.2 Philosophy of Mind and Social Cognition

**Dennett's Intentional Stance:** Dennett (1987) provides the conceptual foundation. I apply his insight about intentionality as instrumental stance to strategic interaction.

**Pinker on Common Knowledge:** Pinker (2025) provides the mechanism linking intentionality to social coordination. His analysis of the Emperor's New Clothes dynamic and relational models integration is central to the social immunity concept.

**Fiske's Relational Models:** Fiske (1991) identifies four fundamental relational structures. AIT uses this framework to diagnose relational mismatch when Level 1 agents (operating only in Market Pricing) interact with Level 3 agents (capable of all four modes).

## 7.3 Legal Theory

**Law and Economics:** Posner (1973) and the Chicago school model legal actors as rational utility maximizers. AIT extends this to heterogeneous agents with different cognitive architectures and social enforcement susceptibilities.

**Behavioral Law and Economics:** Jolls, Sunstein, and Thaler (1998) incorporate psychological findings. AIT identifies a different problem: not universal biases, but asymmetric cognitive architectures between humans and institutions, compounded by asymmetric social enforcement susceptibility.

**Corporate Personhood:** The debate over corporate legal personality has a long genealogy. Dewey (1926) famously argued that 'person' is simply whatever the law chooses to treat as a rights-bearing unit, a position AIT radicalizes: legal personhood creates entities whose intentionality level diverges systematically from the humans who interact with them. More recently, Orts (2013) and Smith (2015) have explored how corporate structure shapes moral agency. AIT contributes a specific mechanism: corporate structure selects for Level 1 optimization regardless of the intentionality levels of individual employees.

**AI and Legal Responsibility:** The emerging literature on AI liability grapples with similar problems. Scherer (2016) identifies the 'opacity problem': we cannot inspect AI decision processes to assign fault. Vladeck (2014) proposes strict liability frameworks. Solum (2019) has explored questions of legal personhood for artificial agents, asking whether and when AI systems might bear legal responsibility. AIT offers a complementary perspective: the relevant question is not whether AI can achieve personhood, but how to design legal frameworks that function correctly when one party operates at Level 0-1 while the other operates at Level 3.

The answer, as developed in Section 6, points toward outcome-based liability and common knowledge forcing mechanisms rather than attribution of intent.

## 8. Limitations and Future Directions

### 8.1 Limitations

**Operationalization Challenges:** Assigning intentionality levels to real-world actors involves judgment calls. The framework's value depends on developing reliable measurement protocols.

**Parameter Calibration:** The Dennett-Nash Gap calculations depend on parameters calibrated from simulations. Empirical measurement remains an open problem.

**Alternative Explanations:** The case studies are consistent with AIT but don't rule out alternative explanations. AIT offers one lens, not the only one.

**Social Immunity Measurement:** The concept of social immunity is theoretically grounded but difficult to measure directly. Developing proxies for social enforcement susceptibility is necessary for empirical validation.

### 8.2 Future Directions

**Experimental Validation:** Design laboratory experiments measuring Dennett-Nash Gaps and social immunity effects. Manipulate apparent intentionality level and social enforcement context.

**Common Knowledge Intervention Design:** Develop and test mechanisms that create common knowledge among dispersed parties to break the Emperor's New Clothes dynamic.

**Legal Implementation:** Develop operationalizable legal tests for intentionality mismatch and social immunity exploitation.

**Corporate Criminal Liability Regimes:** The OECD Anti-Bribery Convention provides a natural experiment for AIT validation. Forty-one countries have implemented nearly identical legal templates requiring corporate mens rea, compliance programs, and spontaneous reporting incentives. Argentina's Ley 27401 (2017) is one instance; the UK Bribery Act 2010, US FCPA, and France's Loi Sapin II are others. Systematic comparison of enforcement outcomes across jurisdictions could test the prediction that Level 3 legal categories applied to Level 1 entities produce compliance theater regardless of local legal culture. Preliminary analysis suggests that concepts like 'corporate tolerance' and 'spontaneous denunciation' generate predictable gaps between formal compliance and actual corruption reduction.

**AI Governance:** As AI systems assume greater roles in consequential decisions, the intentionality mismatch problem will intensify. Current AI ethics frameworks largely assume that articulating Level 3 principles (fairness, dignity, transparency) will somehow constrain Level 0-1 systems. AIT predicts this approach will fail absent enforcement mechanisms that do not presuppose common knowledge capacity. Research should explore outcome-based AI liability, mandatory algorithmic auditing, and 'common knowledge forcing' mechanisms that enable coordinated response to AI harms despite the atomization of affected individuals.

## 9. Conclusion

This paper has developed Asymmetric Intentionality Theory, integrating Dennett's intentional stance framework with Pinker's common knowledge analysis to explain systematic disadvantages faced by high-intentionality players in legal games. The framework identifies two mechanisms: the intentionality cost (wasted cognitive effort modeling non-existent beliefs) and social immunity (escape from shame, reputation, and reciprocity constraints that require common knowledge capacity).

The case studies suggest substantial gaps: in tech ToS, users may lose 15-48% of potential welfare. In Argentine constitutional politics, legislators lose approximately 40% of their power. In sovereign debt negotiations, Argentina lost 40 percentage points of face value. These estimates are imprecise but point toward effects worth investigating.

The Extended Phenotype connection offers a candidate evolutionary mechanism: legal personhood creates selection pressure for Level 1 optimization and social immunity. The social immunity mechanism helps explain why AI ethics frameworks disappoint: they address Level 3 concerns through mechanisms that presuppose common knowledge capacity, while AI systems operate at Level 1 with structural immunity to social enforcement.

Game theory has assumed homogeneous rationality for 70 years. But many important strategic interactions of the 21st century involve asymmetric intentionality and asymmetric social enforcement susceptibility. If so, classical game theory is a special case. The general case requires something closer to AIT.

Future research should apply the AIT framework systematically to corporate criminal liability regimes globally. The OECD Anti-Bribery Convention template, now implemented in 41 countries, provides a natural experiment: identical Level 3 legal architecture (mens rea requirements, compliance programs, spontaneous reporting incentives) applied to Level 1 entities across diverse legal traditions. Preliminary analysis of Argentina's Ley 27401 suggests that concepts like 'corporate tolerance' and 'spontaneous denunciation' are ontologically incoherent when applied to entities incapable of genuine intentionality. The resulting 'compliance theater', extensive documentation of ethical commitments with minimal behavioral change, may represent an extended phenotype optimized for regulatory survival rather than corruption prevention.

## References

Aumann, R. J. (1999). Interactive epistemology I: Knowledge. *International Journal of Game Theory*, 28(3), 263-300.

Baron-Cohen, S., Leslie, A. M., & Frith, U. (1985). Does the autistic child have a 'theory of mind'? *Cognition*, 21(1), 37-46.

Camerer, C. F. (2003). *Behavioral game theory: Experiments in strategic interaction*. Princeton University Press.

Chalmers, D. J. (1996). *The conscious mind: In search of a fundamental theory*. Oxford University Press.

Dawkins, R. (1982). *The extended phenotype: The long reach of the gene*. Oxford University Press.

Dennett, D. C. (1987). *The intentional stance*. MIT Press.

Dennett, D. C. (1995). *Darwin's dangerous idea: Evolution and the meanings of life*. Simon & Schuster.

Dewey, J. (1926). The historic background of corporate legal personality. *Yale Law Journal*, 35(6), 655-673.

Fiske, A. P. (1991). *Structures of social life: The four elementary forms of human relations*. Free Press.

Jolls, C., Sunstein, C. R., & Thaler, R. (1998). A behavioral approach to law and economics. *Stanford Law Review*, 50, 1471-1550.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263-291.

Kinderman, P., Dunbar, R., & Bentall, R. P. (1998). Theory-of-mind deficits and causal attributions. *British Journal of Psychology*, 89(2), 191-204.

Obar, J. A., & Oeldorf-Hirsch, A. (2020). The biggest lie on the Internet: Ignoring the privacy policies and terms of service policies of social networking services. *Information, Communication & Society*, 23(1), 128-147.

OECD (2016). *The liability of legal persons for foreign bribery: A stocktaking report*. OECD Publishing.

Orts, E. W. (2013). *Business persons: A legal theory of the firm*. Oxford University Press.

Pinker, S. (2025). *When Everyone Knows That Everyone Knows: Common Knowledge and the Science of Harmony, Hypocrisy and Outrage*. [Forthcoming].

Posner, R. A. (1973). *Economic analysis of law*. Little, Brown.

Scherer, M. U. (2016). Regulating artificial intelligence systems: Risks, challenges, competencies, and strategies. *Harvard Journal of Law & Technology*, 29(2), 353-400.

Searle, J. R. (1980). Minds, brains, and programs. *Behavioral and Brain Sciences*, 3(3), 417-424.

Simon, H. A. (1955). A behavioral model of rational choice. *Quarterly Journal of Economics*, 69(1), 99-118.

Solum, L. B. (2019). Artificial meaning. *Washington Law Review*, 89, 69-86.

Vladeck, D. C. (2014). Machines without principals: Liability rules and artificial intelligence. *Washington Law Review*, 89, 117-150.

# Appendix A: Formal Arguments

## A.1 Existence Argument (Proposition 1)

**Claim:** Every finite asymmetric intentionality game possesses at least one AIE.

**Argument:** Consider game $G = (P, S, u, I, M)$ with effective payoff function $u^*(p, s) = u(p, s) - \tau(I(p)) - \beta \cdot \text{Mismatch}(p) + \sigma(p)$. For a given strategy profile s, $\tau(I(p))$, Mismatch(p), and $\sigma(p)$ are constants. Thus $u^*(p, \cdot)$ differs from $u(p, \cdot)$ only by a player-specific constant. Best responses under $u^*$ are identical to best responses under u shifted by constants. Standard existence theorems guarantee existence.

## A.2 Common Knowledge Threshold Argument

**Claim:** Common knowledge capacity requires intentionality at Level 3 or above.

**Argument:** Common knowledge of proposition P between agents A and B requires at minimum: A knows that B knows that A knows P. This is third-order belief attribution: A believes (about B's belief (about A's belief (about P))). By definition, this requires Level 3 intentionality. Level 2 agents can model first-order beliefs of others ('B believes P') but not second-order beliefs ('B believes A believes P').

## A.3 Social Enforcement Threshold Argument

**Claim:** Social enforcement mechanisms require common knowledge capacity in the target agent.

**Argument:** Consider shame. Shame modifies behavior when: (1) agent knows it violated norm N; (2) agent knows others know of violation; (3) agent knows others know agent knows of violation; (4) agent knows others expect agent to feel bad about (3). This is Level 4 reasoning. The argument can be weakened to Level 3 if we treat (4) as implicit in social context. Either way, Level 2 or below is insufficient. Similar arguments apply to reputation (requiring modeling of future others' models of self) and reciprocity (requiring tracking of mutual expectation structures over time).

# Appendix B: Computational Implementation

The ToS game simulation uses the following parameters: Service value $V = 80$; Privacy damage coefficient $D = 60$; Compliance cost coefficient = 50; Intentionality cost constants $c_1 = 0.1$, $c_2 = 0.1$; Mismatch sensitivity $\beta = 10$; Social immunity premium $\sigma = 2.90$.

Algorithm: Fixed-point iteration with social immunity component. Initialize user threshold $t = 0.5$. Platform best-responds by choosing aggressiveness a maximizing profit subject to $a \leq t$, with social immunity premium added to platform payoff. User updates threshold based on effective payoff including expected social enforcement (which never materializes). Iterate until convergence.

Results: Equilibrium aggressiveness $a^* \approx 0.735$. Platform effective payoff $\approx 15.80$. User (Level 3) effective payoff $\approx -21.40$. Counterfactual Level 1 user $\approx +18.20$. Dennett-Nash Gap $\approx 37.20$ (including social immunity premium of 2.90).

Sensitivity: Varying $\sigma \in [0, 5]$ yields gaps from 34.30 to 42.20. The social immunity premium contributes 7-12% of total gap across parameter ranges.

# Appendix C: Measurement Protocol

Operationalizing intentionality levels requires mapping Dennett's philosophical framework to observable behaviors:

**Level 0:** Behavior predictable from input-output mappings without goal attribution.

**Level 1:** Consistent goal pursuit across varied contexts; no evidence of second-order modeling.

**Level 2:** Behavior changes based on inferred opponent beliefs; evidence of theory of mind.

**Level 3:** Strategic deception, reputation management, complex social coordination; susceptibility to shame and reciprocity expectations.

For social immunity assessment, test responsiveness to: (1) public shaming campaigns; (2) reputation damage not correlated with measurable metrics; (3) implicit reciprocity expectations. Immunity is indicated by responsiveness only to measurable effects, not to the social mechanisms themselves.